AD 502 – Machine Learning

Pre-Reading Material – Lecture 31

Neural Network-II

Architecture and Learning of Neural Networks

The architecture of a neural network describes how its neurons are organized into layers and connected to one another. The architecture determines how information is processed to produce an output.

Layers of a Neural Network

A typical feedforward neural network consists of three types of layers:

1. Input Layer

The input layer receives the data supplied to the network. For example, in a student-performance prediction system, inputs may include attendance and previous examination marks.

2. Hidden Layer

Hidden layers process input information and learn useful patterns. A neural network may contain one or more hidden layers.

3. Output Layer

The output layer generates the final prediction or classification. For example, it may predict whether a student is likely to pass or fail.

Activation Functions

An activation function determines the output of a neuron based on its processed input. It enables neural networks to learn complex relationships.

Common activation functions include:

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- Sigmoid: Produces outputs between 0 and 1.
- Tanh: Produces outputs between −1 and 1.
- ReLU: Produces zero for negative inputs and retains positive inputs.
- Softmax: Converts output scores into probabilities for multiclass classification.

How Does a Neural Network Learn?

Neural networks learn by repeatedly processing training data and adjusting their parameters to reduce prediction errors.

The main stages are:

1. Forward propagation: The network generates a prediction from the input data.
2. Loss calculation: The prediction is compared with the expected output.
3. Backpropagation: The network determines how its parameters contribute to the prediction error.
4. Parameter updating: Weights and biases are adjusted using an optimization algorithm.
5. Iteration: The process repeats to improve the model's performance.

Training and Overfitting

Training data helps the network learn patterns, while validation data supports model selection. Test data evaluates performance on unseen examples.

Overfitting occurs when a neural network learns the training data too closely and performs poorly on new data.

Conclusion: Neural network architecture, activation functions and training procedures work together to enable a model to learn meaningful patterns from data.

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